Replication information for
"Why have CEO pay levels become less diverse?"

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1. Replication procedure

To replicate the tables and figures in the article and in the Internet Appendix, researchers will need access to data from the original data sources (see section 3 of this document for a list of data vendors). The provided Stata/Python/SAS code combines data from these sources to create the variables used in the analysis.

There are two parts for the replication process:

1. The preparation of data and construction of variables used in the analysis. This involves steps 1-8 in section 1.1.
2. The execution of the code that generates the figures and tables in the article and in the Internet Appendix; this occurs in step 9 in section 1.2.

To be able to run the code and generate the article's findings (i.e., to execute step 9) without running the preparatory code in steps 1-8, we also provide *randomized pseudo data*. Given that this is however pseudo data, step 9 will *not* reproduce the article's findings; instead, it just illustrates the execution of the code provided for step 9.

1.1 Data preparation

If you have access to the original datasets from the vendors (see section 3), please follow the steps in the order outlined below to replicate the results in the article and in the Internet Appendix:

1. Run SAS file `prepare-stock-return-volatility.sas`
2. Run **Step 1** of Stata file `prepare-data.do`
3. Run Stata file `prepare-iss-peergroups.do`
4. Run Stata file `prepare-issInfluence.do`
5. Run **Steps 2-15** of Stata file `prepare-data.do`

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6. Run the following Python files to compute network measures:
   - For network measures at the economy-level: `prepare-networkAnalysis-EconomyLevel.py`
   - For network measures at the industry-level: `prepare-networkAnalysis-ÌndustryLevel.py`
   - For network measures at the firm-level: `prepare-networkAnalysis-PeergroupLevel.py`
7. Run **Steps 16-32** of Stata file `prepare-data.do`
8. Run Stata file `simulations.do`

   ⇒ At this point all the input files for the analysis in `analysis.do` have been created.

1.2 Data analysis

To generate the figures and tables in the article and in the Internet Appendix, follow the final step. **This step can be executed directly with the supplied pseudo data.**

9. Run Stata file `analysis.do`. This will generate the following output:
   – Manuscript: Figures 1-7, Tables 1-10
   – Internet Appendix: Figures IA.1-IA.9, Tables IA.1-IA.6

2. Description of replication files

- `prepare-stock return and volatility.sas`: This file computes annual firm-level market value of equity, stock return, total and idiosyncratic volatility.
- `prepare-issInfluence.do`: This file computes for each firm the fraction of its shares owned by institutional investors who are overwhelmingly following ISS recommendations in their proxy voting.
- `prepare-iss-peergroups.do`: This file computes for each firm and year the ranked potential peers that ISS would assign following its disclosed industry- and size-based peer group methodology (see Appendix B for details).
- `prepare-networkAnalysis-EconomyLevel.py`,
  `prepare-networkAnalysis-IndustryLevel.py`,
  `prepare-networkAnalysis-PeergroupLevel.py`:

  These files compute annual network-based clustering measures (*reciprocity*, *transitivity*, *clustering coefficient*, *triangles*) for directed graphs at various levels. All network-based clustering measures are computed using the "networkx" Python library.

  – File `-EconomyLevel.py` computes these measures annually for an *economy-wide* directed graph containing all compensation peer connections in a given year.

  – File `-IndustryLevel.py` computes these measures annually for each GICS2-industry / each GICS4-industry / each GICS6-industry. That is, the directed graph consists of a given *industry* network of compensation peer connections.

- File `-PeergroupLevel.py` computes these network-based clustering measures annually for each firm within its *compensation peer group*. Each graph includes the directed connections of the main firm to its compensation peers as well as the directed connections of the peers among themselves (based on their respective peer groups).

- `prepare-simulations.do`: This file simulates the consequences to economy-wide CEO pay dispersion if firms were to start selecting their compensation peers following ISS's peer group methodology, and then setting their pay to some given target percentile of such peer groups.

- `prepare-data.do`: This file creates the main data files used in the file `analysis.do`.

- `analysis.do`: This is the main Stata file that runs the analysis and generates the output – i.e., the figures and tables – of the article.

3. Data sources used in the analysis

- `Compustat`: fundamentals
- `CRSP`: stock file; CCM linktable; company header history; indices; delistings
- `Execucomp`: CEO pay data
- `ISS Executive Compensation Analytics`: CEO pay data
- `ISS IncentiveLab`: Compensation peer groups; compensation consultant data
- `ISS Voting Analytics`: Shareholder voting data
- `ISS RiskMetrics`: directors file; governance file
- `ISS mutual fund voting`
- `AuditAnalytics`: Compensation peer group data; CEO pay data
- `Equilar`: Compensation peer group data; CEO pay data
- `NRG metrics`: CEO pay data
- `BoardEx`: CEO pay data
- `GMI`: CEO pay data
- `Capital IQ People Intelligence`: Compensation data
- `Refinitiv`: Institutional 13F holdings
- `Russell 3000` historical constituents
- `CRSP mutual fund holdings`
- `Brian Bushee`: Institutional investor classification data (https://accounting-faculty.wharton.upenn.edu/bushee/)
- `Federal Reserve bank of Minneapolis`: Consumer price index (https://www.minneapolisfed.org/about-us/monetary-policy/inflation-calculator/consumer-price-index-1913-)